

Austin Lin

<https://austlin.github.io> | austinlin35@gmail.com | New London, CT 06320

U.S. Citizen | Open to Relocation

TECHNICAL SKILLS

Languages: Python, Java

Tools: MATLAB/Simulink, ROS2/Gazebo

EXPERIENCE

General Dynamics Electric Boat

Electrical Systems Engineer, New London, CT

Sept 2024 - Present

- Implemented a new shock modeling method for electrical cables, reducing excess cable length in a space-constrained environment by up to 60% compared to existing methods. Used MATLAB to implement cost-constraint optimization and statistical methods to fuse data from authoritative sources and modeling.
- Generated interactive reports to deliver modeling results using MATLAB's App Designer.
- Led a team of engineers to continuously implement ship class-wide design changes of cognizant electrical systems to improve safety, reliability, and maintenance.
- Overcame production bottlenecks by providing engineering assistance to other disciplines (Design, Work Planning, Deckplate/Trades), maintaining schedule velocity.

L3Harris C5 Systems

Systems Engineering Intern, Camden, NJ

June 2023 - Aug 2023

- Demonstrated product quality to the customer by performing System Operability Verification Tests for MarCom IVCS and Symphony. Completed CompTIA Network+ Training.
- Performed root-cause troubleshooting on network/radio configurations with Linux-based hardware.

UMD Space Systems Lab

Undergraduate Researcher, College Park, MD

May 2022 - Aug 2022

- Delivered spacecraft robot arm testing and modeling results to facilitate In-Space Servicing mission requirements. Conducted research with Benjamin Reed from Quantum Space and Dr. David Akin.
- Calculated and visualized the workspace of a robotic arm to reach payloads for in-orbit servicing. Used Denavit-Hartenberg parameters to calculate forward kinematics.
- Produced and executed a test procedure to validate Python model results for 24+ spacecraft configurations. Designed and fabricated a 1/4-scale physical robotic arm mockup.

PROJECTS

Battery Simulation and State of Charge Estimator

- Created an interactive Python/Shiny app to visualize the impact of adjusting Luenberger observer gain and measurement noise on battery state of charge estimation accuracy and convergence.

Robotics and Autonomous Systems

- Implemented a maze-solving algorithm on a robot using LIDAR for live localization and mapping. Used Gazebo and ROS2 to model a virtual environment to tune the algorithm before field testing.
- Programmed robots to identify, grab, and hand off LEGO bricks to each other using computer vision. Trained a YOLOv8 model for object identification and odometry, and used ZeroMQ through TCP/IP for communication.
- Developed hardware and software for a robot arm and controller. Programmed C++ firmware for an Arduino microcontroller to handle serial communication to a computer that calculates and graphically displays arm kinematics.

EDUCATION

University of Maryland, College Park, MD

Aug 2020 - Aug 2024

B.S. Mechanical Engineering, Minor in Robotics and Autonomous Systems

Scholars Public Leadership Program